

BE 306: Genetic Engineering: principles and applications

Course Code	BE 306
Course Name	Genetic Engineering: principles and applications
L-T-P-C	3-1-0-4
Prerequisites	IC136 or consent of the faculty member
Intended for	UG (IDD Bioengg, 3rd year)
Distribution	Discipline core
Mutual Exclusion	None
Approval	50th BoA (Jul 2023)

Course Contents

Introduction: Gene and mRNA structure and properties; analysis of DNA and RNA sequences, DNA and RNA modifying enzymes (Restriction Enzymes, DNA ligase, Klenow enzyme, T4 DNA polymerase, Polynucleotide kinase, Alkaline phosphatase). Introduction to genetic engineering and GMOs. [12 hours]

PCR and Its Applications : Primer, Primer designing, Thermostable DNA polymerases, PCR, Types of PCR – multiplex, nested, reverse transcriptase, cDNA synthesis, real-time PCR, touchdown PCR, hot start PCR, colony PCR. Site-directed mutagenesis, Mutation detection, PCR in molecular diagnostics, Viral and bacterial detection. [11 hours]

Cloning Vectors: Bacterial and viral based plasmids (PUC19, Bluescript vectors, M13 vectors, SV-40 vectors, Phagemids, Cosmids); Artificial chromosome vectors (YACs; BACs); Plant based vectors, Ti and Ri as vectors, Selection of vectors, Expression vectors (pMal; GST; pET-based vectors) [8 hours]

Cloning Methodologies: Restriction Enzyme Based Cloning; PCR Cloning (TOPO or TA); Ligation Independent Cloning (LIC); Seamless Cloning (SC); Recombinational Cloning; Gibson Assembly (Isothermal Assembly Reaction); Expression cloning, Construction of genomic and cDNA libraries. [12 hours]

Introduction of DNA into cells: Introduction of DNA into bacterial cells (transformation methods), viruses (transduction methods), mammalian cells (Transfection techniques), plant tissues (Transfection techniques, particle bombardment), and model organisms (microinjections). [7 hours]

Genetic engineering applications, case studies and ethical issues: Recent developments in genetic engineering methods; Applications of genetic engineering in agriculture and medicine; GMOs and GEMs; Socio-economic, cultural, and ethical issues. [6 hours]

Text Books

1. Terry A. Brown, **Gene Cloning: An Introduction**, 8th Edition, Wiley-Blackwell, 2021.
2. Sandy B. Primrose, Richard Twyman, **Principles of Gene Manipulation and Genomics**, 8th edition, John Wiley Blackwell, 2016.

References

1. Michael R. Green and J. Sambrook, **Molecular Cloning: A Laboratory Manual**, 4th Edition, Vols 1-3, Cold Spring Harbor Laboratory Press, CSHL, 2012.
2. B. Alberts, R. Heald, A. Johnson, D. Morgan, M. Raff, **Molecular Biology of the Cell**, 7th Edition, W.W. Norton & Co Inc, 2022.
3. Relevant research articles/reviews will be advised relating to the topic being taught.